

Post-Processing Adjustment (Decision Fixes)

This step focuses on adjusting decisions after the model has already produced an output. It is used as a practical mitigation approach when changing the data or retraining the model is not immediately feasible. This stage applies when harm is concentrated at the decision threshold or when urgent mitigation is required.

This stage must only be applied when causal fairness analysis identifies bias at the decision stage or when immediate mitigation is required.

When to Use

Apply the Post-Processing Toolkit when:

- Retraining the model is slow, costly, or operationally infeasible
- Immediate mitigation is required to reduce harm or unfair outcomes
- The model is locked, third-party, or cannot be modified due to technical or contractual constraints

This step is often used as a temporary or risk-containment measure rather than a permanent solution.

When Not to Use

Post-processing should not be used as a primary solution when:

- Root causes of bias are known and correctable
- Model retraining is feasible
- The intervention risks masking systemic bias

Post-processing must not replace upstream correction in high-risk systems.

Allowed Actions

Teams may implement the following decision-level interventions:

- Threshold adjustments to align decision cutoffs across groups or reduce unequal rejection or approval rates
- Reject-option logic near uncertainty, where borderline cases are handled more cautiously or routed for human review instead of automated decisions

These actions directly influence outcomes without altering model internals.

Limitations

Post-processing does not address the root causes of bias in data or model behavior. As a result, these interventions:

- Require continuous monitoring to detect drift or unintended effects
- Should be reviewed periodically and replaced with upstream fixes when feasible

Governance Requirement

In high-risk systems, post-processing must:

- Be explicitly justified as a temporary or constrained intervention
- Be accompanied by a plan for upstream correction
- Be approved through governance review

Failure to address root causes must be formally documented.

Required Output

Every post-processing intervention must include:

- A Threshold Change Summary documenting what was adjusted and why
- A Before/After Outcome Comparison showing the impact on fairness metrics and affected groups

Clear documentation ensures transparency, accountability, and future review readiness. Post-processing should not replace upstream correction when retraining is feasible.

Connection to Intervention Selection and Validation

Post-processing is selected when causal analysis identifies bias at the decision stage or when immediate mitigation is required.

All post-processing interventions must be validated to ensure:

- Fairness improvement at the decision level
- No unintended harm or drift
- Stability over time

Post-processing must be continuously monitored and replaced with upstream fixes when feasible.

Post-processing directly influences decision outcomes and is most effective as a controlled, temporary mitigation strategy rather than a long-term solution.